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Borrow joints from wood: how Stonehenge's stones lock together

The free guide to the reel on Stonehenge's joints: how mortise-and-tenon and tongue-and-groove joints hold the sarsen circle, the numbers behind the stones, and a clay-and-cardboard way to make a locking lintel.



A stone circle with carpenter's joints

The sarsen circle at Stonehenge is radiocarbon dated to roughly 2600 to 2400 BC. Its lintels are held in place on the uprights with mortise-and-tenon joints, a feature the article on Stonehenge describes as unique among contemporary monuments. The lintels are fitted to one another with tongue-and-groove joints, and the trilithons are linked by complex jointing.

These are woodworking joints, cut in stone: a knob (tenon) on top of each upright fits into a socket (mortise) in the underside of the lintel.

Why joints and not cement

A lintel resting on two uprights is held down by its own weight, but a sideways push could slide it off. A knob in a socket stops that sliding, and a tongue in a groove keeps neighbouring lintels in line. In the circle, they make a continuous ring that behaves as one piece.

The numbers

Standing sarsens	About 4 m high, 2.1 m wide and 1 m deep; each weighs around 25 tons.
Trilithon stones	Up to 50 tons each.
Lintel top height	About 4.9 m above the ground.
Distance	The sarsens came from West Woods, about 25 km away; the smaller bluestones (about 2 tons each) from the Preseli Hills, 230 to 240 km away.

Try it yourself: a locking lintel

1. Roll two cylinders of modelling clay (the uprights) and press a small rounded knob on top of each.
2. Flatten a longer piece of clay (the lintel) and press two shallow holes into its underside with your thumb, spaced to match the knobs.
3. Place the lintel on the uprights without the knobs and push it sideways: it slides.

4 Lower it onto the knobs and push sideways again: it resists. That is the mortise and tenon.

5 Make two lintels and add a tongue and groove between their ends: see how they stay aligned.

The rule to remember

If a joint works in one material, ask whether it works in another.

Default answer	Stack and rely on weight.
Form-finder answer	Cut a joint that stops the one movement that would break it.
Check	Ask: which direction could this fail in, and what shape would stop exactly that?

Go further

Stonehenge (Wikipedia)
https://en.wikipedia.org/wiki/Stonehenge

Mortise and tenon (Wikipedia)
https://en.wikipedia.org/wiki/Mortise_and_tenon

English Heritage: Stonehenge
https://www.english-heritage.org.uk/visit/places/stonehenge/

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Facts: Wikipedia (Stonehenge).

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